

Calculus and Ordinary Differential Equations

AS1109 – Section H

Accelerated Student Notes: Lessons 4–8

How to use this handout. Definitions, formulas, representative examples and the difficult diagrams are already printed. Use the blank spaces for short explanations, calculations, questions and extra examples from class. You are not expected to copy the printed material from the board.

Lesson	Main content
4	Continuity; first partial derivatives; geometric and physical meaning
5	Higher-order partial derivatives; Mixed Derivative Theorem
6	Implicit differentiation; Laplace equation; wave equation
7	Differentiability; tangent planes; linearisation
8	Differentials; sensitivity; absolute/relative/percentage error

This sequence follows Thomas–Finney §§12.2–12.4. In this edition the Mixed Derivative Theorem occurs in §12.3, immediately after second-order partial derivatives, so we study it there.

Lesson 4: Continuity and first partial derivatives

Learning goals. You should be able to:

- test continuity at a point;
- state where standard formulas are continuous;
- calculate f_x and f_y ;
- interpret partial derivatives as local slopes/rates of change.

1. Continuity at a point

A function $f(x, y)$ is **continuous at** (a, b) if

- (i) $f(a, b)$ is defined;
- (ii) $\lim_{(x,y) \rightarrow (a,b)} f(x, y)$ exists;
- (iii) $\lim_{(x,y) \rightarrow (a,b)} f(x, y) = f(a, b)$.

Polynomials are continuous everywhere. Rational functions are continuous where their denominators are non-zero. Standard sums, products and compositions of continuous functions are continuous wherever the resulting formula is defined.

$$f(x, y) = \begin{cases} \frac{x^2 y}{x^2 + y^2}, & (x, y) \neq (0, 0), \\ 0, & (x, y) = (0, 0) \end{cases}$$

is continuous at $(0, 0)$ because the limit is 0 and equals $f(0, 0)$.

2. Partial derivatives

When differentiating with respect to one variable, hold the other independent variables fixed.

$$f_x(a, b) = \lim_{h \rightarrow 0} \frac{f(a + h, b) - f(a, b)}{h}, \quad f_y(a, b) = \lim_{h \rightarrow 0} \frac{f(a, b + h) - f(a, b)}{h}.$$

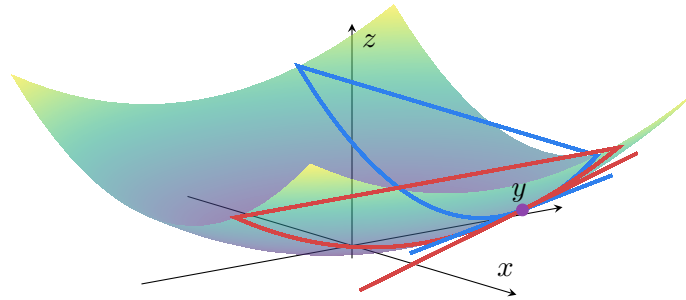
If

$$f(x, y) = x^2 y + 3xy^2,$$

then

$$f_x = 2xy + 3y^2, \quad f_y = x^2 + 6xy.$$

3. Geometric meaning: two trace slopes



At the marked point, the blue tangent line gives the x -direction slope f_x (with y fixed), while the red tangent line gives the y -direction slope f_y (with x fixed).

If $T(x, y)$ is temperature on a plate, T_x and T_y are local temperature changes per unit displacement in the coordinate directions. Their signs give the direction of increase/decrease; their magnitudes compare steepness.

Exercises for Lesson 4

1. State where $\frac{x+y}{x-y}$ is continuous.
 2. Determine whether the piecewise function above is continuous at the origin.
 3. Find f_x, f_y for $f = e^{xy} + \sin(x+y)$.
 4. Find f_x, f_y for $f = \ln(x^2 + y^2)$ and state where they apply.
 5. For $T = 100 - x^2 - 2y^2$, find $T_x(1, 2)$ and $T_y(1, 2)$ and interpret the signs.
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Lesson 5: Higher-order partial derivatives and mixed derivatives

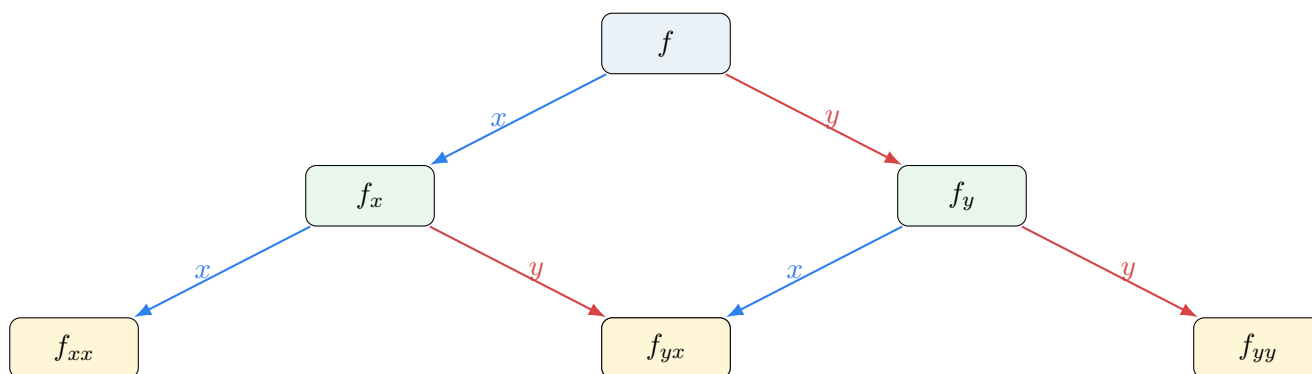
Learning goals. You should be able to calculate all second partial derivatives, read the order of subscripts correctly, and state when $f_{xy} = f_{yx}$ is guaranteed.

1. Four second partial derivatives

Once f_x and f_y are known, differentiate again:

$$f_{xx}, \quad f_{xy}, \quad f_{yx}, \quad f_{yy}.$$

The rightmost subscript records the derivative taken first.



Let

$$f = x^3y^2 + e^{xy}.$$

Then

$$f_x = 3x^2y^2 + ye^{xy}, \quad f_y = 2x^3y + xe^{xy},$$

so

$$\begin{aligned} f_{xx} &= 6xy^2 + y^2e^{xy}, \\ f_{xy} &= 6x^2y + e^{xy} + xye^{xy}, \\ f_{yx} &= 6x^2y + e^{xy} + xye^{xy}, \\ f_{yy} &= 2x^3 + x^2e^{xy}. \end{aligned}$$

2. Mixed Derivative Theorem

If f_x, f_y, f_{xy}, f_{yx} are defined throughout an open region containing (a, b) and the mixed partials are continuous at (a, b) , then

$$f_{xy}(a, b) = f_{yx}(a, b).$$

Do not memorise “mixed partials are always equal.” Equality requires suitable smoothness. There are functions for which both mixed partials exist at a point but are unequal.

Exercises for Lesson 5

1. Find all four second partial derivatives of $f = x^2y + xy^2$.
 2. Find f_{xy} and f_{yx} for $f = e^{xy}$.
 3. For $f = \ln(x^2 + y^2)$, compute f_{xx} and f_{yy} where defined.
 4. Explain why equality of mixed partials is expected for an ordinary polynomial model.
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Lesson 6: Implicit differentiation and two important PDE models

Learning goals. You should be able to differentiate an implicit relation involving $z = z(x, y)$ and recognise the Laplace and wave equations together with their physical meanings.

1. Implicit differentiation

If

$$F(x, y, z) = 0$$

defines $z = z(x, y)$, then differentiating with respect to x gives

$$F_x + F_z z_x = 0,$$

so, when $F_z \neq 0$,

$$z_x = -\frac{F_x}{F_z}.$$

Similarly,

$$z_y = -\frac{F_y}{F_z}.$$

For

$$x^2 + y^2 + z^2 = 9,$$

$$z_x = -\frac{x}{z}, \quad z_y = -\frac{y}{z}.$$

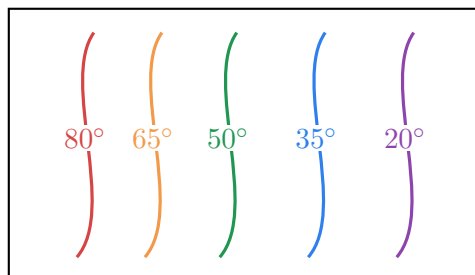
At $(1, 2, 2)$:

$$z_x = -\frac{1}{2}, \quad z_y = -1.$$

2. Laplace equation

$$u_{xx} + u_{yy} = 0$$

is the two-dimensional Laplace equation.



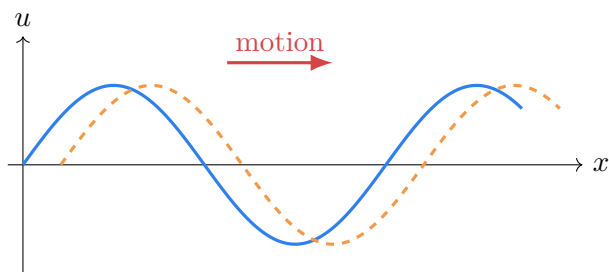
steady-state temperature contours

A steady temperature field in a thin plate with no internal heat source is a standard engineering setting in which Laplace's equation appears. The same equation also models electrostatic potential in source-free regions.

3. Wave equation

$$u_{tt} = c^2 u_{xx}$$

is the one-dimensional wave equation; c is the wave speed.



The wave equation models small transverse vibrations of a stretched string and analogous wave propagation in acoustics and elastic systems.

Exercises for Lesson 6

1. If $x^2 + yz + z^2 = 5$, find z_x .
 2. If $xyz + x + y + z = 1$, find formulas for z_x and z_y where possible.
 3. Verify that $u = x^2 - y^2$ satisfies $u_{xx} + u_{yy} = 0$.
 4. If $u(x, t) = \sin(x - ct)$, verify directly that it satisfies $u_{tt} = c^2 u_{xx}$.
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Lesson 7: Differentiability, tangent planes and linearisation

Learning goals. You should be able to explain differentiability as a good local planar approximation, use the continuous-partial test, and construct the linearisation of a function of two variables.

1. Differentiability

For small $\Delta x, \Delta y$, a differentiable function satisfies

$$\Delta f = f_x(a, b)\Delta x + f_y(a, b)\Delta y + \text{smaller error}.$$

A convenient sufficient condition used in this course is: if f_x and f_y are continuous in an open region containing (a, b) , then f is differentiable at (a, b) .

Differentiability implies continuity. Existence of f_x and f_y alone does not guarantee differentiability.

2. Linearisation

At (a, b) the standard linear approximation is

$$L(x, y) = f(a, b) + f_x(a, b)(x - a) + f_y(a, b)(y - b).$$

For $f(x, y) = \sqrt{x^2 + y^2}$ at $(3, 4)$,

$$f(3, 4) = 5, \quad f_x(3, 4) = \frac{3}{5}, \quad f_y(3, 4) = \frac{4}{5}.$$

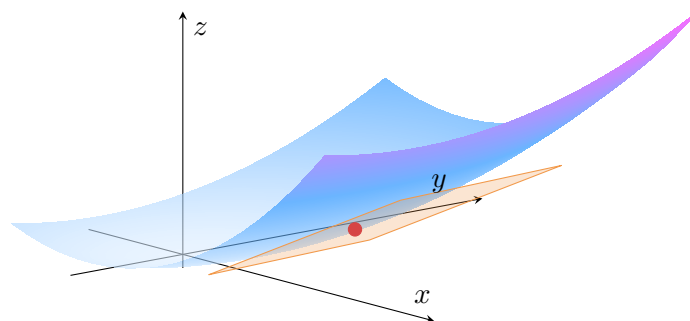
Hence

$$L(x, y) = 5 + \frac{3}{5}(x - 3) + \frac{4}{5}(y - 4).$$

Therefore

$$f(3.02, 3.97) \approx 4.988.$$

3. The tangent plane is the graph of the linearisation



Close to the marked point, the curved surface is well approximated by the tangent plane. This is why a nonlinear engineering model can often be treated locally as a linear model.

Exercises for Lesson 7

1. Find the linearisation of $f = x^2 + xy + y^2$ at $(1, 1)$.
 2. Use it to estimate $f(1.02, 0.99)$.
 3. Find the linearisation of $f = e^{x+y}$ at $(0, 0)$.
 4. Explain why a function with continuous first partial derivatives near a point is particularly convenient for engineering approximation.
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Lesson 8: Differentials, sensitivity and error estimation

Learning goals. You should be able to predict small output changes using differentials and estimate absolute, relative and percentage error.

1. Total differential

For $z = f(x, y)$,

$$dz = df = f_x dx + f_y dy.$$

For sufficiently small input changes,

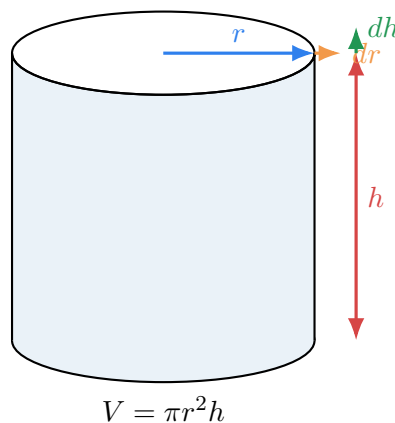
$$\Delta f \approx df.$$

2. Engineering example: cylindrical tank

For

$$V = \pi r^2 h,$$

$$dV = 2\pi r h dr + \pi r^2 dh.$$



$$V = \pi r^2 h$$

At $r = 5$, $h = 12$, $dr = 0.02$, $dh = -0.03$,

$$dV = 2\pi(5)(12)(0.02) + \pi(25)(-0.03) = 1.65\pi \approx 5.18.$$

3. Relative and percentage change

For a non-zero quantity f ,

$$\text{relative change} \approx \frac{df}{f}, \quad \text{percentage change} \approx 100 \frac{df}{f} \%.$$

For the cylinder,

$$\frac{dV}{V} = 2 \frac{dr}{r} + \frac{dh}{h}.$$

With the data above,

$$\frac{dV}{V} = 0.0055,$$

so the predicted percentage change is approximately 0.55%.

4. Measurement-error example

If $A = xy$ with $x = 40$ cm, $y = 25$ cm and

$$|dx| \leq 0.1, \quad |dy| \leq 0.1,$$

then

$$|dA| \leq |y||dx| + |x||dy| \leq 25(0.1) + 40(0.1) = 6.5 \text{ cm}^2.$$

Since $A = 1000 \text{ cm}^2$, this is approximately a 0.65% relative error.

Differentials give first-order estimates, not exact changes. They are most accurate when the input changes are small and the function is smooth near the operating point.

Exercises for Lesson 8

1. For $V = \pi r^2 h$, derive dV/V .
 2. A cylinder has $r = 4$, $h = 10$ with possible errors $|dr| \leq 0.01$, $|dh| \leq 0.03$. Estimate the maximum percentage error in V .
 3. If $R = V/I$, find dR and dR/R .
 4. For $P = xy^2$, find dP/P in terms of dx/x and dy/y .
 5. Explain which measurement should receive tighter tolerance when its corresponding sensitivity coefficient is much larger.
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Checklist before the next block

You should now be able to say “yes” to the following:

- I can connect a two-variable limit with continuity.
- I can compute and interpret first partial derivatives.
- I can compute pure and mixed second partial derivatives.
- I know that equality of mixed partials requires suitable hypotheses.
- I can differentiate an equation implicitly to obtain z_x or z_y .
- I can identify the Laplace and wave equations and name standard physical applications.
- I can construct a tangent-plane/linear approximation.
- I can use differentials for sensitivity and first-order error estimates.

Next: multivariable Chain Rule, implicit differentiation via the Chain Rule, extrema and saddle points.